

Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



Year 10 2026

Test 1: Review Topics

Test: Tuesday 6 October, Period 6 — 37 marks, 40 minutes, the whole unit

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Numbers

- ☐ Convert an 8-bit binary number to denary and back; add two 8-bit binary numbers, showing your working, and recognise when the result overflows 8 bits.
- ☐ Convert a 2-digit hexadecimal number to denary, and explain the method.
- ☐ Perform a left binary shift, and state what it does to the value (multiplies by a power of two).
- ☐ Work out the smallest and largest denary number representable in a given number of bits, and the number of bits needed to represent a given range of values (e.g. enough characters, or enough colours).

2. Characters

- ☐ Explain how a character set works: every character has a unique binary code, and **ASCII** uses 8 bits per character.
- ☐ Use the fact that a character set is **logically ordered** to work out one character's code from another's (e.g. if F is 70, work out L).

3. Images

- ☐ State what **colour depth** means, and calculate the maximum number of colours representable at a given bit depth (2^{bits}).
- ☐ Decode a binary bit-stream into a pixel grid, using the colour depth to work out how many bits belong to each pixel, and name the colour each group of bits represents.
- ☐ State what **metadata** means for an image file, and give examples (width, height, colour depth, file type).
- ☐ Explain two effects of raising an image's colour depth (larger file size; more colours can be represented).

4. Sound

- ☐ State what **sample rate** means for a sound recording (how often the sound wave is measured).
- ☐ Explain how **bit depth** affects a sound recording (how many possible values each sample can take) and what happens to the file if you change it.

5. Compression

- ☐ Explain the difference between **lossy** and **lossless** compression.
- ☐ Identify the best type of compression for a given file (text or image) and **justify** the choice — text needs every character preserved exactly (lossless); an image can lose some detail without it being noticeable (lossy).

Remember: maximum colours = 2^{bits} file size grows with both resolution and colour depth lossless keeps everything, lossy throws some away

Practise with the Data Representation Review Packet on your class page.